

MOLECULAR PHYLOGENETICS & PHYLOGENOMICS — MOCK EXAM 1

A total of 27 closed questions, each worth one point, with no penalty for incorrect answers.

01. Phylogenetic systematics (cladistics) groups organisms based primarily on:

- A. Overall morphological and phenotypic similarity.
- B. Their unique shared ecological niches.
- C. Their evolutionary history and shared ancestry relationships.
- D. The total percentage of identity calculated across whole genomes.

02. A shared derived character state that is inherited from the immediate most recent common ancestor of a focal clade and defines monophyly is a/an:

- A. Sympleisiomorphy.
- B. Synapomorphy.
- C. Autapomorphy.
- D. Homoplasy.

03. Choose the INCORRECT statement regarding the Unweighted Pair Group Method with Arithmetic Mean (UPGMA):

- A. It calculates a pairwise distance matrix to cluster sequences.
- B. It produces unrooted, additive trees with varying lineage speeds.
- C. It explicitly assumes a constant molecular clock over evolutionary time.
- D. It results in an ultrametric tree where all tips are equidistant from the root.

04. Which of the following represents an empirical model specifically optimized for tracking amino acid substitution patterns?

- A. General Time Reversible (GTR)
- B. Kimura 2-Parameter (K80)
- C. Jones-Taylor-Thornton (JTT)
- D. Jukes-Cantor (JC69)

05. In Metropolis-Coupled Markov Chain Monte Carlo (MC3), the role of the 'heated chains' is to:

- A. Perform the final parameter and branch length posterior calculations.
- B. Flatten the posterior probability landscape to cross deep valleys and escape local optima.
- C. Artificially alter base frequencies to match composition variations.
- D. Accelerate the burn-in phase by dropping early iterations dynamically.

06. In an alignment scoring system, a/an _____ gap penalty applies a constant linear cost per gap position, while a/an _____ gap penalty charges a high cost to initiate a gap and a lower cost to extend it.

07. Which branch support metric is explicitly designed as a continuous variant of the standard bootstrap to handle massive taxon samplings without being heavily penalized by unstable rogue tips?

- A. Site Concordance Factor (sCF)
- B. Approximate Likelihood Ratio Test (aLRT)
- C. Transfer Bootstrap Expectation (TBE)
- D. Posterior Probability (PP)

08. An outgroup is included a priori in a phylogenetic analysis primarily to:

- A. Increase the signal-to-noise ratio of variable markers.
- B. Introduce alternative model testing environments.
- C. Provide an evolutionary direction and root to the target tree.
- D. Saturate the alignment matrix to prevent Long Branch Attraction.

09. When summarizing a Bayesian posterior distribution of trees into a single interpretable topology, the Maximum Clade Credibility (MCC) tree is defined as:

- A. The consensus tree displaying all clades that appear in more than 50% of the sample pool.
- B. The single real sampled tree from the MCMC run that maximizes the product of posterior clade probabilities.
- C. An artificial topology assembled by merging the longest branch lengths from across distinct chains.
- D. A tree optimized to minimize the absolute sum of all internal node branch variances.

10. Which of the following is NOT a major advantage of using transcriptomes over whole genomes for phylogenomics?

- A. It yields a highly expanded set of genetic markers compared to targeted single genes.
- B. It is significantly cheaper, allowing for higher taxon sampling capacity.
- C. It provides flawless coverage of non-coding genomic regions and structural mutations.
- D. It functions effectively without the mandatory requirement of a pre-existing reference genome.

11. Under a continuous trait evolution model driven by Brownian Motion (BM), the absolute variance of a quantitative trait across a group of species increases:

- A. Exponentially relative to the number of nodes.
- B. Linearly as a function of time since divergence.
- C. Logarithmically until it reaches a stable adaptive peak.
- D. Symmetrically based on the strength of selection towards the optimum.

12. Which of the following tree rearrangement search algorithms explores the largest landscape of alternative topologies, offering the lowest risk of getting trapped in local optima?

- A. Nearest Neighbor Interchange (NNI)
- B. Subtree Pruning and Regrafting (SPR)
- C. Tree Bisection and Reconnection (TBR)
- D. Metropolis-Hastings Swapping (MHS)

13. A polyphyletic taxonomic group is uniquely defined as an assemblage that:

- A. Includes an ancestral lineage and all of its descendant branches without exception.
- B. Consists of an ancestor and some, but not all, of its descendant lineages.
- C. Includes two or more separate groups, each derived from separate common ancestors.
- D. Contains exclusively terminal taxa displaying identical autapomorphies.

14. What is the primary technical limitation of using a simple Reciprocal Best Hit (RBH) approach to identify orthologous genes between genomes?

- A. It requires massive computational clusters and week-long runtimes.
- B. It is strictly limited to mitochondrial marker sets.
- C. It fails completely to detect one-to-many or many-to-many orthology relationships.
- D. It is highly sensitive to the choice of transition/transversion ratios.

15. Under-partitioning a complex genomic supermatrix dataset during Maximum Likelihood tree construction typically leads to:

- A. Overfitting of random background sequencing noise.
- B. Biased parameter estimations due to forcing distinct evolutionary rates into a single model.
- C. An inflation of the Effective Sample Size (ESS) metrics.
- D. The automatic elimination of parsimony-informative positions.

16. What size of Effective Sample Size (ESS) for likelihood parameters is widely recognized as the minimum required threshold to confirm adequate mixing in an MCMC run?

- A. 10
- B. 50
- C. 100
- D. 200

17. If Pagel's lambda (λ) metric for a given Phenotypic trait is calculated to be close to 0, this explicitly indicates that:

- A. The trait perfectly matches pure Brownian Motion expectations structured by the phylogeny.
- B. The trait distribution is completely independent of the phylogeny and resembles random noise.
- C. The trait is under extreme stabilizing selection pulling it towards a tight adaptive peak.
- D. The trait has experienced significant horizontal gene transfer across distant lineages.

18. Which of the following is NOT an approach utilized to estimate divergence times in molecular clock analyses?

- A. Least Squares Dating (LSD)
- B. Penalized Maximum Likelihood (PML)
- C. Bayesian Inference using BEAST
- D. Nearest Neighbor Interchange (NNI)

19. In codon substitution models, an estimated ratio of _____ indicates the action of purifying selection, whereas a ratio of _____ explicitly indicates positive selection.

20. In a Maximum Likelihood framework, the total likelihood of a phylogenetic tree is calculated as the product of individual site probabilities, an approach that mathematically assumes that:

- A. All sites evolve at a completely identical substitution rate over time.
- B. Each individual site in the sequence alignment evolves completely independently.
- C. Transitions and transversions occur with equal probabilities across the tree.
- D. Base frequencies remain strictly stationary across all ancestral nodes.

21. The Shimodaira-Hasegawa (SH) topology test is designed to compare multiple trees and is more _____ than the Kishino-Hasegawa (KH) test because it explicitly adjusts for comparisons.
22. In the context of fossil calibration, a/an _____ group contains the last common ancestor of all living members of a clade and all its descendants, while a/an _____ group contains extinct lineages that are outside the crown but closer to it than to any other living group.
23. In Bayesian inference, the probability that the evolutionary model and tree topology are correct given the observed sequence alignment data is known as the:
- A. Prior Probability.
 - B. Likelihood function.
 - C. Posterior Probability.
 - D. Marginal Likelihood.
24. In the coalescent framework, the probability that two gene copies merge in the immediate preceding generation is inversely proportional to the _____, and one standard coalescent unit represents exactly _____ generations for haploid elements.
25. An alignment column is classified as a 'singleton' if it contains a variant character state present in only _____ taxon, meaning this specific position is completely _____ for Maximum Parsimony analysis.
26. The phenomenon where the absolute evolutionary rate at a specific alignment position changes across different branches of a phylogenetic tree over time is known as _____, and failing to model it can lead to severe directional errors called _____.
27. Which of the following describes the behavior of disruptive selection on a quantitative trait within a population?
- A. The average value of the trait shifts towards one extreme phenotype.
 - B. Individuals with extreme phenotypes on both ends of the curve exhibit the highest fitness, creating a bimodal distribution.
 - C. Extreme phenotypes are heavily selected against, reducing trait variance around the mean.
 - D. Mutations accumulate purely randomly without altering phenotypic fitness profiles.

A SINGLE OPEN-ENDED QUESTION WORTH 5 POINTS, ANSWER IN MAX. THREE SENTENCES:

Explain the biological phenomenon of Incomplete Lineage Sorting (ILS) and explicitly state the two primary population or evolutionary conditions under which it is most likely to cause gene tree discordance.
